

Bowen Yang

Center of Mathematical Sciences and Applications
Department of Mathematics
Harvard University
Cambridge, MA 02138
bowen_yang@g.harvard.edu

Research Interests

Algebra and mathematical physics, particularly commutative algebra in exactly solvable quantum lattice models. Homotopy-theoretic structure of quantum cellular automata and topological phases of matter. Quantum information and error-correction.

Employment

Postdoctoral Fellow, Center of Mathematical Sciences and Applications, Harvard University 2024–present

Lecturer, Department of Mathematics, Harvard University Spring 2027

Teaching Fellow, Department of Mathematics, Harvard University Spring 2026

Publications

Published and Accepted

1. A mathematical theory of topological invariants of quantum spin systems (with Adam Artymowicz, Anton Kapustin), **Commun. Amer. Math. Soc.** *accepted*, arXiv:2410.19287.
2. Transition between 2D Symmetry Protected Topological Phases on a Klein Bottle (with Vibhu Ravindran, Xie Chen), **Phys. Rev. B** (2026) arXiv:2510.00587.
3. Clifford quantum cellular automata from topological quantum field theories and invertible subalgebras (with Meng Sun, Zongyuan Wang, Nathanan Tantivasadakarn, Yu-An Chen), **PRX Quantum** (2026) arXiv:2509.07099.
4. Categorifying Clifford QCA, **Commun. Math. Phys.** (2026), arXiv:2504.14811.
5. Homological invariants of Pauli stabilizer codes (with Blazej Ruba), **Commun. Math. Phys.** (2024). arXiv:2204.06023
6. Spatial decay of Kubo’s canonical correlation functions, **Lett. Math. Phys.** (2021). arXiv:1912.10831
7. A classification of invertible phases of bosonic quantum lattice systems in one dimension (with Anton Kapustin, Nikita Sopenko), **J. Math. Phys.** (2021). arXiv:2012.15491
8. Toy Teichmüller spaces of real dimension 2: the pentagon and the punctured triangle (with Yudong Chen et al.), **Geom. Dedicata** (2018). arXiv:1704.04331

9. “Strange” combinatorial quantum modular forms (with Amanda Folsom et al.), **J. Number Theory** (2017).

Preprints

1. K-Theoretic Obstructions to Linearizing QCA Representations (with Mattie Ji), arXiv:2606.19657.
2. Resolving Spurious Topological Entanglement Entropy in Stabilizer Codes (with P. Han, Z. Liang, Y. Wang, Y. Gu, Y. Chen), arXiv:2604.24847.
3. The Classification of Pauli Stabilizer Codes: A Lattice and Continuum Treatise (with Matthew Yu), arXiv:2604.24847.
4. Quantum Cellular Automata: The Group, the Space, and the Spectrum (with Mattie Ji), arXiv:2602.16572.
5. Witt Groups and Bulk-Boundary Correspondence for Stabilizer States (with Blazej Ruba), arXiv:2509.10418.
6. Operator algebra and algorithmic construction of boundaries and defects in (2+1)D topological Pauli stabilizer codes (with Zijian Liang, Joseph T. Iosue, Yu-An Chen), arXiv:2410.11942 (2024).

Invited Talks since Fall 2024

- 2026, **Stanford University**, *Quantum Cellular Automata Workshop*
- 2026, **Massachusetts Institute of Technology**, *Infinite-Dimensional Algebra Seminar*
- 2026, **La Jolla Meeting**, *Simons Collaboration on Global Categorical Symmetries*
- 2025, **University of Colorado Boulder**, *Topology Seminar*
- 2025, **Harvard University**, *Freedman Seminar*
- 2025, **New York University**, *Simons Collaboration on Global Categorical Symmetries*
- 2025, **SLMath**, *Workshop on Infinite-Dimensional Division Algebras*
- 2025, **Massachusetts Institute of Technology**, *Algebraic Topology Seminar*
- 2025, **University of Pennsylvania**, *Mathematical Physics Seminar*
- 2025, **University of California, Davis**, *Lattice Seminar* (online)
- 2024, **Amherst College**, *Colloquium*
- 2024, **Institute for Advanced Study**, *CMP/QFT Group Meeting*

Education

Ph.D. in Mathematics, California Institute of Technology Advisor: Anton Kapustin	2018–2023
B.A. in Mathematics, <i>summa cum laude</i> , Amherst College	2014–2018
Visiting Master’s Student, New College, University of Oxford	2016–2017

Teaching Experience

Senior Thesis Advisor, Harvard University Student: Luke Zhu	Summer 2026–Spring 2027
Summer REU Mentor, Harvard University Student: Vincent Costa	Summer 2026
Instructor, Harvard University Math 21B — Linear Algebra and Ordinary Differential Equations	Spring 2026
Instructor, California Institute of Technology Math 8 — Calculus for Problem Solving	Fall 2021
Course Assistant, California Institute of Technology Math 2/102 — Differential Equations Math 3/103 — Introduction to Probability and Statistics Math 5/105a — Introduction to Abstract Algebra Math 5/105b — Introduction to Abstract Algebra Math 5/105c — Introduction to Abstract Algebra Math 110c — Harmonic Analysis Math 121a — Combinatorial Analysis Math 121b — Combinatorial Analysis Math 1c — Calculus Math 120a — Commutative Algebra	2018–2022 Fall 2018 Winter 2019 Fall 2019 Winter 2020 Spring 2019 Spring 2020 Winter 2021 Spring 2021 Spring 2022 Fall 2022
Teaching Assistant/ Math Fellow, Amherst College CS 101 — Python, Spring 2015 CS 121 — Java, Fall 2015 Math 350 — Groups, Rings, and Fields Math 450 — Measure Theory and Integration Math 355 — Introduction to Analysis Math 405 — Lie Algebras	2015–2018 Fall 2015 Spring 2016 Fall 2017 Fall 2017
Counselor, Ross Mathematics Program	Summers 2017, 2019

Awards and Fellowships

Scott Russell Johnson Graduate Dissertation Prize	2023
---	------

Rufus B. Kellogg University Fellowship (\$30,000 per annum)

2018–2021

Amherst International Student Scholarship

2014–2018

Professional Service

Referee for *Communications in Mathematical Physics*; *Quantum Computing Theory in Practice (QCTiP)*; *Qualitative Theory of Dynamical Systems*.